

## 5.3 Irreducible Polynomials and Unique Factorization

Every integer  $n \geq 2$  is a product of primes, and Chapter 0 remarked that the convention “1 is not prime” is what makes such factorizations unique. Polynomials over a field behave the same way, with irreducible polynomials in the role of primes and non-zero constants in the role of  $\pm 1$ . This section defines irreducible polynomials, proves Euclid’s lemma from the gcd machinery of the previous section, and proves that every non-constant polynomial factors into irreducibles in exactly one way. The catch is that “irreducible” depends on the field: the same polynomial may factor over  $\mathbb{R}$  but not over  $\mathbb{Q}$ , or over  $\mathbb{F}_5$  but not over  $\mathbb{F}_3$ . So we also collect tools for deciding irreducibility over specific fields: roots in low degree, the Rational Root Theorem, and Eisenstein’s criterion. Throughout,  $F$  is a field.

### 5.3.1 Irreducible polynomials

For integers, the building blocks are the primes: integers  $p \geq 2$  whose only positive divisors are 1 and  $p$  (Definition 0.2.2). To transplant this to  $F[x]$ , recall from the first section that every non-zero constant divides every polynomial, and that multiplying by a non-zero constant changes nothing for divisibility. So the trivial divisors of  $p$  are the non-zero constants and the associates  $cp$ . A building block should have no others.

*An irreducible polynomial is a non-constant polynomial that cannot be split into two factors of smaller degree.*

#### Definition 5.3.1 — Irreducible Polynomial

Let  $F$  be a field. A polynomial  $p \in F[x]$  is **irreducible over  $F$**  if

- (1)  $p$  is **non-constant**, that is,  $\deg p \geq 1$ ; and
- (2) whenever  $p = gh$  with  $g, h \in F[x]$ , **one of  $g, h$**  is a constant.

A non-constant polynomial that is not irreducible is **reducible over  $F$** .

In words: condition (1) excludes 0 and the non-zero constants. Condition (2) says every factorization of  $p$  is trivial: one factor is a constant, which is non-zero because  $p \neq 0$ , and the other is then an associate of  $p$ . The phrase **over  $F$**  is part of the notion, since  $g$  and  $h$  must lie in  $F[x]$ .

Since degrees add (Theorem 0.8.3), a factorization  $p = gh$  with  $p \neq 0$  has  $\deg g + \deg h = \deg p$ . So  $g$  is constant exactly when  $\deg h = \deg p$ . This gives a working form of the definition: **a non-constant  $p$  is reducible if and only if  $p = gh$  with  $1 \leq \deg g < \deg p$  and  $1 \leq \deg h < \deg p$ .**

#### Example. (First Irreducible and Reducible Polynomials)

- (a) Show that every polynomial of degree 1 is irreducible over every field.
- (b) Show that the constants 0 and 2 are not irreducible over  $\mathbb{Q}$ .

- (c) Show that  $x^2 + x$  is reducible over every field, and that  $x^3$  is reducible over every field.

*Solution.*

- (a) If  $\deg p = 1$  and  $p = gh$ , then  $\deg g + \deg h = 1$  with both degrees in  $\mathbb{N}$ , so one of them is 0: one factor is a constant. Condition (1) holds since  $\deg p = 1$ . Hence  $p$  is irreducible, whatever the field. For example  $x$ ,  $2x + 4$  and  $x - \sqrt{2}$  (over  $\mathbb{R}$ ) are irreducible.
- (b) Both are constants, so condition (1) fails. They are the degenerate cases, and they matter: 2 divides everything in  $\mathbb{Q}[x]$ , like 1 among the integers, and it is not a building block.
- (c)  $x^2 + x = x(x + 1)$  with both factors of degree 1, and  $x^3 = x \cdot x^2$  with factors of degrees 1 and 2. In each case both factors are non-constant, so condition (2) fails.

Why insist on “non-constant”? For the same reason 1 is not a prime. If non-zero constants were allowed as building blocks, factorizations could never be unique:  $x = 2 \cdot \frac{1}{2}x = 2 \cdot 3 \cdot \frac{1}{6}x = \dots$  over  $\mathbb{Q}$ , with any number of constant factors. The factorization theorem below pulls out one constant in front and uses only non-constant, monic irreducibles.

For polynomials of degree 2 and 3, irreducibility is decided by roots alone.

### Theorem 5.3.1 — Irreducibility in Degree 2 and 3

Let  $F$  be a field and  $p \in F[x]$ .

- (a)  $p$  has a factor of degree 1 in  $F[x]$  if and only if  $p$  has a root in  $F$ . In particular, if  $\deg p \geq 2$  and  $p$  has a root in  $F$ , then  $p$  is reducible over  $F$ .
- (b) If  $\deg p \in \{2, 3\}$ , then  $p$  is irreducible over  $F$  if and only if  $p$  has **no** root in  $F$ .

*Proof.*

- (a) ( $\Leftarrow$ ) If  $c \in F$  is a root, then  $p = (x - c)q$  by Theorem 5.1.3 (b). ( $\Rightarrow$ ) If  $p = (ax + b)q$  with  $a \neq 0$ , then  $c = -a^{-1}b$  satisfies  $ac + b = 0$ , so  $p(c) = 0 \cdot q(c) = 0$  by Theorem 0.8.5. For the last sentence, if  $\deg p \geq 2$  and  $p = (x - c)q$ , then  $\deg q = \deg p - 1 \geq 1$ , so both factors are non-constant of degree less than  $\deg p$ , and  $p$  is reducible.
- (b) We prove both directions by contraposition. ( $\Rightarrow$ ) If  $p$  has a root, it is reducible by (a), since  $\deg p \geq 2$ . ( $\Leftarrow$ ) Suppose  $p$  is reducible. By the working form of the definition,  $p = gh$  with  $\deg g, \deg h \geq 1$  and  $\deg g + \deg h = \deg p \in \{2, 3\}$ . Two integers that are at least 1 and sum to 2 or 3 cannot both be at least 2, so one of  $g, h$  has degree 1. By (a),  $p$  has a root in  $F$ . This proves the theorem. ■

Now the standard examples, each checked with the theorem.

- $x^2 + 1$  **over  $\mathbb{R}$  and over  $\mathbb{C}$** . For real  $c$ ,  $c^2 + 1 \geq 1 > 0$ , so there is no real root, and  $x^2 + 1$  is irreducible over  $\mathbb{R}$  (and over  $\mathbb{Q}$ ). Over  $\mathbb{C}$ ,  $x^2 + 1 = (x - i)(x + i)$  is reducible.

- $x^2 - 2$  **over  $\mathbb{Q}$  and over  $\mathbb{R}$** . A rational root would be a rational number with square 2, which does not exist (Theorem 0.2.3). So  $x^2 - 2$  is irreducible over  $\mathbb{Q}$ . Over  $\mathbb{R}$ ,  $x^2 - 2 = (x - \sqrt{2})(x + \sqrt{2})$ .
- $x^2 + x + 1$  **over  $\mathbb{F}_2$** . The values at 0 and 1 are 1 and  $1 + 1 + 1 = 1$ . No root, so it is irreducible over  $\mathbb{F}_2$ . It is the only irreducible polynomial of degree 2 over  $\mathbb{F}_2$  (Exercise B3).

The non-example by minimal change is  $x^2 + x$  over  $\mathbb{F}_2$ : change the constant term of  $x^2 + x + 1$  from 1 to 0, and 0 becomes a root, so  $x^2 + x = x(x + 1)$  is reducible. What fails is condition (2).

*Warning.* “No roots” implies “irreducible” **only in degrees 2 and 3**. The polynomial  $(x^2 + 1)^2 = x^4 + 2x^2 + 1$  has no real root, since its values are at least 1, but it is reducible over  $\mathbb{R}$ : it is a product of two factors of degree 2. In degree 4 and higher, a polynomial can split into non-linear factors without having any root.

*Quick check.* Over which of the fields  $\mathbb{Q}$ ,  $\mathbb{Q}(\sqrt{2})$  (Example ( $\mathbb{Q}(\sqrt{2})$  Is a Field)),  $\mathbb{R}$  is  $x^2 - 2$  irreducible? Is  $x^2 + 1$  irreducible over  $\mathbb{F}_5$ ?

*Solution.* Over  $\mathbb{Q}$  only. The field  $\mathbb{Q}(\sqrt{2})$  contains  $\sqrt{2}$ , a root of  $x^2 - 2$ , so  $x^2 - 2 = (x - \sqrt{2})(x + \sqrt{2})$  is reducible there, and likewise over  $\mathbb{R}$ ; over  $\mathbb{Q}$  it is irreducible, as shown above. Over  $\mathbb{F}_5$ ,  $2^2 + 1 = 5 = 0$ , so 2 is a root and  $x^2 + 1$  is reducible by Theorem 5.3.1: indeed  $x^2 + 1 = (x - 2)(x + 2)$  in  $\mathbb{F}_5[x]$ .

## 5.3.2 Euclid’s lemma

What makes primes useful is not their definition but a property the definition implies: if a prime divides a product, it divides one of the factors. For polynomials this follows quickly from the previous section. The first step is that an irreducible polynomial has only two possible gcds with anything.

### Lemma 5.3.2 — The gcd with an Irreducible Polynomial

Let  $F$  be a field,  $p \in F[x]$  irreducible over  $F$ , and  $f \in F[x]$ . Then either  $p \mid f$  or  $\gcd(p, f) = 1$ .

*Proof.* Let  $d = \gcd(p, f)$ . Since  $p \neq 0$ ,  $d$  is monic, and  $d \mid p$  by Theorem 5.2.3 (a), say  $p = du$ . Since  $p$  is irreducible,  $d$  or  $u$  is a constant. If  $d$  is a constant, then  $d = 1$ , since  $d$  is monic. If  $u$  is a constant, it is non-zero, and  $p = ud$  divides every multiple of  $d$ ; since  $d \mid f$  by Theorem 5.2.3 (a), Proposition 5.1.2 (a) gives  $p \mid f$ . This proves the lemma. ■

### Theorem 5.3.3 — Euclid’s Lemma for Polynomials

Let  $F$  be a field and  $p \in F[x]$  irreducible over  $F$ . If  $q, r \in F[x]$  and  $p \mid qr$ , then  $p \mid q$  or  $p \mid r$ .

*Proof.* Suppose  $p \nmid q$ . By Lemma 5.3.2,  $\gcd(p, q) = 1$ . Since  $p \mid qr$ , Theorem 5.2.8 gives  $p \mid r$ . This proves the theorem. ■

The hypothesis “irreducible” is essential. The polynomial  $x^2$  divides  $x \cdot x$  but divides neither factor; the lemma does not apply because  $x^2 = x \cdot x$  is reducible. Over  $\mathbb{F}_2$ ,  $x^2 + 1 = (x + 1)^2$  divides  $(x + 1)(x + 1)$  but not  $x + 1$ , for the same reason.

For unique factorization we need the lemma for products of any length, and we need to know what “divides” means between two monic irreducibles.

### Corollary 5.3.4 — Euclid’s Lemma for Several Factors

Let  $F$  be a field and  $p \in F[x]$  irreducible over  $F$ .

- (a) If  $q_1, \dots, q_k \in F[x]$  with  $k \geq 1$  and  $p \mid q_1 q_2 \cdots q_k$ , then  $p \mid q_i$  for some  $i$ .
- (b) If  $q$  is irreducible over  $F$ ,  $p \mid q$ , and  $p, q$  are both monic, then  $p = q$ .

*Proof.*

- (a) We use induction on  $k$ . For  $k = 1$  there is nothing to prove. Let  $k \geq 2$  and assume the statement for  $k - 1$  factors. Since  $p \mid (q_1 \cdots q_{k-1}) q_k$ , Theorem 5.3.3 gives  $p \mid q_1 \cdots q_{k-1}$  or  $p \mid q_k$ . In the first case the induction hypothesis gives  $p \mid q_i$  for some  $i \leq k - 1$ . Either way the statement holds for  $k$ .
- (b) Write  $q = pu$ . Since  $q$  is irreducible,  $p$  or  $u$  is constant. But  $p$  is not constant, being irreducible, so  $u$  is a non-zero constant. Comparing leading coefficients of the monic polynomials  $q$  and  $p$  gives  $u = 1$ , so  $p = q$ . ■

## 5.3.3 Unique factorization

Here is the theorem that Chapter 0’s remark about primes was pointing towards, now for polynomials. The constant in front is forced to be the leading coefficient; the rest is a product of monic irreducibles, unique up to order.

### Theorem 5.3.5 — Unique Factorization in $F[x]$

Let  $F$  be a field and  $f \in F[x]$  non-constant.

- (a) **Existence.** There are a non-zero constant  $c \in F$ , an integer  $k \geq 1$ , and monic polynomials  $p_1, \dots, p_k \in F[x]$ , each irreducible over  $F$ , with

$$f = c p_1 p_2 \cdots p_k.$$

- (b) **Uniqueness.** If also  $f = c' q_1 q_2 \cdots q_l$  with  $c' \in F$  non-zero,  $l \geq 1$ , and  $q_1, \dots, q_l$  monic and irreducible over  $F$ , then  $c = c'$ ,  $k = l$ , and after renumbering the  $q_j$  we

have  $p_i = q_i$  for every  $i$ .

In both factorizations, the constant is the leading coefficient of  $f$ .

*Idea. Existence:* if  $f$  is irreducible, stop; otherwise split it into two factors of smaller degree and factor each of them, which is strong induction on the degree. **Uniqueness:** compare leading coefficients to get  $c = c'$ . Then  $p_k$  divides  $q_1 \cdots q_l$ , so by Euclid's lemma it equals some  $q_j$ . Cancel it from both sides; the two shorter factorizations are equal by induction on the number of factors  $k$ . The one thing to watch is that after canceling, the right-hand side must still have at least one factor.

*Proof.* First a remark used in both parts. A product of monic polynomials is monic, by Theorem 0.8.3 and induction on the number of factors. So if  $f = c p_1 \cdots p_k$  with the  $p_i$  monic, then the leading coefficient of  $f$  is  $c \cdot 1 = c$ . This proves the final sentence of the theorem.

- (a) We prove by strong induction on  $n \geq 1$  (Theorem 0.2.10 with  $n_0 = 1$ ) that every  $f$  of degree  $n$  has a factorization as in (a). Let  $\deg f = n$ , assume the statement for all degrees  $1 \leq m < n$ , and let  $c$  be the leading coefficient of  $f$ .

*Case 1:  $f$  is irreducible over  $F$ .* Then  $f = c(c^{-1}f)$ . The polynomial  $c^{-1}f$  is monic, and it is irreducible: it is non-constant, and if  $c^{-1}f = gh$ , then  $f = (cg)h$ , so  $cg$  or  $h$  is constant, hence  $g$  or  $h$  is constant. So  $k = 1$ ,  $p_1 = c^{-1}f$  works.

*Case 2:  $f$  is reducible over  $F$ .* By the working form of the definition,  $f = gh$  with  $1 \leq \deg g < n$  and  $1 \leq \deg h < n$ . By the induction hypothesis,  $g = a p_1 \cdots p_s$  and  $h = b p_{s+1} \cdots p_{s+t}$  with  $a, b$  non-zero constants,  $s, t \geq 1$ , and all  $p_i$  monic irreducible. Then  $f = (ab) p_1 \cdots p_{s+t}$ , with  $ab \neq 0$  by Theorem 0.7.1. This is a factorization as in (a), with  $k = s + t$ .

- (b) By the first remark,  $c$  and  $c'$  are both the leading coefficient of  $f$ , so  $c = c'$ . Multiplying by  $c^{-1}$ , we get  $p_1 \cdots p_k = q_1 \cdots q_l$ . So it suffices to prove, by induction on  $k \geq 1$ , the statement

$U(k)$ : whenever  $p_1 \cdots p_k = q_1 \cdots q_l$  with  $l \geq 1$  and all  $p_i, q_j$  monic and irreducible over  $F$ , we have  $k = l$ , and after renumbering the  $q_j$ ,  $p_i = q_i$  for every  $i$ .

We first note: if a monic irreducible  $q$  equals a product  $r_1 \cdots r_m$  of  $m \geq 2$  monic irreducibles, then  $q = r_1 \cdot (r_2 \cdots r_m)$  with both factors non-constant, since each  $r_i$  has degree at least 1; this contradicts the irreducibility of  $q$ . So **a monic irreducible is never a product of two or more monic irreducibles.** (\*)

*Base case  $k = 1$ .* Then  $p_1 = q_1 \cdots q_l$ . By (\*),  $l = 1$ , and so  $p_1 = q_1$ .

*Inductive step.* Let  $k \geq 2$  and assume  $U(k-1)$ . Suppose  $p_1 \cdots p_k = q_1 \cdots q_l$ . If  $l = 1$ , then  $q_1 = p_1 \cdots p_k$  with  $k \geq 2$ , which is impossible by (\*). So  $l \geq 2$ . Now  $p_k \mid q_1 \cdots q_l$ , so by Corollary 5.3.4 (a),  $p_k \mid q_j$  for some  $j$ , and by Corollary 5.3.4 (b),  $p_k = q_j$ . Renumber the  $q$ 's so that  $j = l$ . Then

$$(p_1 \cdots p_{k-1}) p_k = (q_1 \cdots q_{l-1}) p_k,$$

and since  $p_k \neq 0$ , Corollary 0.8.4 (2) lets us cancel it:  $p_1 \cdots p_{k-1} = q_1 \cdots q_{l-1}$ , with

$k - 1 \geq 1$  and  $l - 1 \geq 1$  factors. By  $U(k - 1)$ ,  $k - 1 = l - 1$ , and after renumbering  $q_1, \dots, q_{l-1}$ ,  $p_i = q_i$  for  $i \leq k - 1$ . Together with  $p_k = q_k$ , this proves  $U(k)$ .

By induction,  $U(k)$  holds for every  $k \geq 1$ , which proves (b). ■

Grouping equal factors, every non-constant  $f \in F[x]$  can be written uniquely as

$$f = c p_1^{e_1} p_2^{e_2} \cdots p_s^{e_s}$$

with  $c$  the leading coefficient,  $p_1, \dots, p_s$  **distinct** monic irreducibles over  $F$ , and exponents  $e_i \geq 1$ . The monic irreducible factors of the characteristic and minimal polynomials of an operator will organize the structure theory of Chapters 8 and 9.

*Warning.* Uniqueness needs the normalization “monic”. Without it, constants can move between factors: over  $\mathbb{Q}$ ,  $2x^2 - 2 = (2x - 2)(x + 1) = (x - 1)(2x + 2) = (\frac{1}{2}x - \frac{1}{2})(4x + 4)$ . All of these are “factorizations into irreducibles”, and they differ only by associates. The theorem removes this freedom by collecting all constants into  $c$ . Uniqueness also refers to a **fixed** field: over  $\mathbb{Q}$  the factorization of  $x^2 - 2$  is  $x^2 - 2$  itself, and over  $\mathbb{R}$  it is  $(x - \sqrt{2})(x + \sqrt{2})$ .

The next example shows the field dependence in one table.

#### Example. (Factoring $x^4 - 1$ Over Six Fields)

Factor  $x^4 - 1$  into a constant times monic irreducibles over  $\mathbb{Q}$ ,  $\mathbb{R}$ ,  $\mathbb{C}$ ,  $\mathbb{F}_2$ ,  $\mathbb{F}_3$ , and, for contrast,  $\mathbb{F}_5$ .

*Solution.* Over every field,  $x^4 - 1 = (x^2 - 1)(x^2 + 1) = (x - 1)(x + 1)(x^2 + 1)$ , by expanding. The factors  $x - 1$  and  $x + 1$  are irreducible by [Example \(First Irreducible and Reducible Polynomials\)](#) (a). Everything depends on  $x^2 + 1$ , which by [Theorem 5.3.1](#) is irreducible exactly when it has no root in the field. The constant in front is always the leading coefficient 1.

- $\mathbb{Q}$  and  $\mathbb{R}$ . For real  $c$ ,  $c^2 + 1 \geq 1$ , so there is no root, and  $x^2 + 1$  is irreducible.
- $\mathbb{C}$ .  $x^2 + 1 = (x - i)(x + i)$ .
- $\mathbb{F}_2$ . Here  $-1 = 1$ , so  $x - 1 = x + 1$ , and  $x^2 + 1 = (x + 1)^2$  because  $2x = 0$ . All four factors coincide.
- $\mathbb{F}_3$ . The values of  $x^2 + 1$  at 0, 1, 2 are 1, 2, 5 = 2. No root, so  $x^2 + 1$  is irreducible. Also  $x - 1 = x + 2$ .
- $\mathbb{F}_5$ .  $2^2 + 1 = 5 = 0$  and  $3^2 + 1 = 10 = 0$ , so  $x^2 + 1 = (x - 2)(x - 3) = (x + 3)(x + 2)$ .

| field          | factorization of $x^4 - 1$ into monic irreducibles |
|----------------|----------------------------------------------------|
| $\mathbb{Q}$   | $(x - 1)(x + 1)(x^2 + 1)$                          |
| $\mathbb{R}$   | $(x - 1)(x + 1)(x^2 + 1)$                          |
| $\mathbb{C}$   | $(x - 1)(x + 1)(x - i)(x + i)$                     |
| $\mathbb{F}_2$ | $(x + 1)^4$                                        |
| $\mathbb{F}_3$ | $(x + 1)(x + 2)(x^2 + 1)$                          |
| $\mathbb{F}_5$ | $(x + 1)(x + 2)(x + 3)(x + 4)$                     |

By [Theorem 5.3.5](#), each row is **the** factorization over that field, up to order. SymPy's `factor_list`, with `extension=I` over  $\mathbb{C}$  and `modulus=p` over  $\mathbb{F}_p$ , returns the same rows. Notice that the numbers of irreducible factors are 3, 3, 4, 4, 3, 4, and that over  $\mathbb{F}_2$  a repeated factor appears although  $x^4 - 1$  has no repeated factor over the other fields.

### 5.3.4 Tools over $\mathbb{Q}$ : rational roots and Eisenstein's criterion

Over  $\mathbb{Q}$ , [Theorem 5.3.1](#) reduces irreducibility of a quadratic or cubic to the question of rational roots. There are infinitely many rational numbers to try, but for integer coefficients only finitely many candidates can work. We write  $\mathbb{Z}[x]$  for the polynomials in  $\mathbb{Q}[x]$  whose coefficients are all integers, as in the first section. Every rational number can be written as  $u/v$  with  $u \in \mathbb{Z}$ ,  $v \geq 1$ , and the only positive integer dividing both  $u$  and  $v$  equal to 1: take  $v \geq 1$  as small as possible, since a common divisor  $e > 1$  would give the representation  $(u/e)/(v/e)$  with a smaller denominator. We say  $u/v$  is in **lowest terms**.

#### Theorem 5.3.6 — Rational Root Theorem

Let  $f = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_0 \in \mathbb{Z}[x]$  with  $n \geq 1$  and  $a_n \neq 0$ . If  $u/v \in \mathbb{Q}$  is in lowest terms and  $f(u/v) = 0$ , then  $u \mid a_0$  and  $v \mid a_n$ .

*Idea.* Clear denominators:  $v^n f(u/v) = a_n u^n + a_{n-1} u^{n-1} v + \cdots + a_0 v^n = 0$ . Every term except the last contains  $u$ , so  $u$  divides  $a_0 v^n$ ; every term except the first contains  $v$ , so  $v$  divides  $a_n u^n$ . Since  $u$  and  $v$  share no factor, the powers of  $v$  can be stripped off one at a time, using Bézout for integers.

*Proof.* Multiplying  $f(u/v) = 0$  by  $v^n$  gives

$$a_n u^n + a_{n-1} u^{n-1} v + \cdots + a_1 u v^{n-1} + a_0 v^n = 0.$$

Hence  $a_0 v^n = -u(a_n u^{n-1} + a_{n-1} u^{n-2} v + \cdots + a_1 v^{n-1})$ , so  $u \mid a_0 v^n$ ; and  $a_n u^n = -v(a_{n-1} u^{n-1} + \cdots + a_0 v^{n-1})$ , so  $v \mid a_n u^n$ .

Since  $v \geq 1$  and the only positive common divisor of  $u$  and  $v$  is 1, Lemma 0.7.2 gives integers  $s, t$  with  $us + vt = 1$ . If  $b \in \mathbb{Z}$  and  $k \geq 1$  with  $u \mid bv^k$ , then

$$bv^{k-1} = bv^{k-1}(us + vt) = u(bv^{k-1}s) + t(bv^k)$$

is a sum of two multiples of  $u$ , so  $u \mid bv^{k-1}$ . Applying this  $n$  times to  $b = a_0$  gives  $u \mid a_0$ . Similarly, if  $v \mid cu^k$  with  $k \geq 1$ , then  $cu^{k-1} = s(cu^k) + v(cu^{k-1}t)$ , so  $v \mid cu^{k-1}$ ; applying this  $n$  times to  $c = a_n$  gives  $v \mid a_n$ . This proves the theorem. ■

A polynomial in  $\mathbb{Q}[x]$  with fractional coefficients can first be multiplied by a common denominator; this does not change its roots or its irreducibility.

**Example. (A Cubic That Is Irreducible Over  $\mathbb{Q}$ )**

Decide whether  $f = 2x^3 - 3x^2 + 4x - 1$  is irreducible over  $\mathbb{Q}$ .

*Solution.* By [Theorem 5.3.6](#), a rational root  $u/v$  in lowest terms has  $u \mid -1$  and  $v \mid 2$  with  $v \geq 1$ . So  $u = \pm 1$  and  $v \in \{1, 2\}$ , and the candidates are  $1, -1, \frac{1}{2}, -\frac{1}{2}$ . Evaluating,

$$f(1) = 2, \quad f(-1) = -10, \quad f\left(\frac{1}{2}\right) = \frac{1}{4} - \frac{3}{4} + 2 - 1 = \frac{1}{2}, \quad f\left(-\frac{1}{2}\right) = -\frac{1}{4} - \frac{3}{4} - 2 - 1 = -4.$$

None is 0, so  $f$  has no rational root. Since  $\deg f = 3$ , [Theorem 5.3.1](#) shows that  $f$  is irreducible over  $\mathbb{Q}$ . (SymPy's `factor_list` over  $\mathbb{Q}$  agrees.)

Beyond degree 3, roots are not enough, as the warning after [Theorem 5.3.1](#) showed. One classical criterion handles many polynomials of any degree at once. It looks at divisibility of the coefficients by a prime.

**Theorem 5.3.7 — Eisenstein's Criterion**

Let  $f = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_0 \in \mathbb{Z}[x]$  with  $n \geq 1$ , and let  $p$  be a prime such that

- (1)  $p \nmid a_n$ ;
- (2)  $p \mid a_i$  for every  $i = 0, 1, \dots, n-1$ ;
- (3)  $p^2 \nmid a_0$ .

Then  $f$  is irreducible over  $\mathbb{Q}$ .

The proof needs one more idea, **Gauss's lemma**: a polynomial with integer coefficients that factors over  $\mathbb{Q}$  already factors over  $\mathbb{Z}$ , with factors of the same degrees. With it, one reduces the coefficients modulo  $p$  and compares. Exercise C2 of this section proves Gauss's lemma and then Eisenstein's criterion, step by step; no result of this chapter depends on the criterion. Here we only use it in examples.

**Example. (Irreducible Polynomials of Every Degree Over  $\mathbb{Q}$ )**

- (a) Show that  $x^n - 2$  is irreducible over  $\mathbb{Q}$  for every  $n \geq 1$ . (b) Show that  $x^4 + 10x + 5$  is irreducible over  $\mathbb{Q}$ .

*Solution.*

- (a) Take  $p = 2$ . The leading coefficient 1 is not divisible by 2; the coefficients  $a_1, \dots, a_{n-1}$  are 0 and  $a_0 = -2$ , all divisible by 2; and  $4 \nmid -2$ . By [Theorem 5.3.7](#),  $x^n - 2$  is irreducible over  $\mathbb{Q}$ . So over  $\mathbb{Q}$  there are irreducible polynomials of every degree. (Over  $\mathbb{R}$ , by contrast,  $x^n - 2$  has the root  $\sqrt[n]{2}$  and is reducible for  $n \geq 2$ .)
- (b) Take  $p = 5$ :  $5 \nmid 1$ ; 5 divides 0, 0, 10, 5; and  $25 \nmid 5$ . So  $x^4 + 10x + 5$  is irreducible over  $\mathbb{Q}$ . Roots alone could not have shown this, since a quartic may factor into two quadratics.

Eisenstein's criterion is sufficient, not necessary. The polynomial  $x^2 + 1$  is irreducible over  $\mathbb{Q}$ , but no prime divides its constant term 1, so condition (2) fails for every  $p$ .

## 5.3.5 Exercises

### 5.3.5.1 A. Check your understanding

#### Exercise A1.

- Define what it means for  $p \in F[x]$  to be irreducible over  $F$ .
- State the Unique Factorization Theorem in  $F[x]$ .
- Determine whether the following statement is true: “a polynomial in  $\mathbb{R}[x]$  of degree at least 1 with no real root is irreducible over  $\mathbb{R}$ .” Justify your answer.
- Is the constant 2 irreducible over  $\mathbb{Q}$ ? Is  $2x + 4$ ?
- Is  $x^2 + 1$  irreducible over  $\mathbb{F}_5$ ? Justify your answer.
- Name a tool that reduces the irreducibility of a cubic in  $\mathbb{Z}[x]$  over  $\mathbb{Q}$  to finitely many evaluations.

*Solution.*

- $p$  is irreducible over  $F$  if  $\deg p \geq 1$  and, whenever  $p = gh$  with  $g, h \in F[x]$ , one of  $g, h$  is a constant ([Definition 5.3.1](#)).
- Every non-constant  $f \in F[x]$  can be written as  $f = cp_1 \cdots p_k$  with  $c \in F$  non-zero and  $p_1, \dots, p_k$  monic and irreducible over  $F$ ; the constant  $c$  is the leading coefficient of  $f$ , and the  $p_i$  are unique up to order ([Theorem 5.3.5](#)).
- False.  $(x^2 + 1)^2$  has no real root, but it is the product of two non-constant factors, so it is reducible over  $\mathbb{R}$ . The statement is true in degrees 2 and 3 only ([Theorem 5.3.1](#)).
- No: 2 is constant, and irreducible polynomials are non-constant by definition. Yes:  $2x + 4$  has degree 1, so it is irreducible ([Example \(First Irreducible and Reducible Polynomials\)](#) (a)).
- No. In  $\mathbb{F}_5$ ,  $2^2 + 1 = 0$ , so  $x^2 + 1 = (x - 2)(x + 2)$  is reducible by [Theorem 5.3.1](#).
- The Rational Root Theorem ([Theorem 5.3.6](#)), combined with [Theorem 5.3.1](#).

### 5.3.5.2 B. Practice

#### Exercise B1: Irreducible or Not?

Determine whether each polynomial is irreducible over each of the given fields. Justify your answers.

- $x^2 + 2$  over  $\mathbb{Q}$ ,  $\mathbb{R}$ ,  $\mathbb{F}_2$  and  $\mathbb{F}_3$ .
- $x^3 + x + 1$  over  $\mathbb{F}_2$  and over  $\mathbb{F}_3$ .
- $x^3 - 3x - 1$  over  $\mathbb{Q}$ .
- $x^4 + 4$  over  $\mathbb{Q}$ .
- $x^5 - 6x + 3$  over  $\mathbb{Q}$ .

*Solution.*

- Over  $\mathbb{Q}$  and  $\mathbb{R}$ : for real  $c$ ,  $c^2 + 2 \geq 2 > 0$ , so there is no root, and  $x^2 + 2$  is

- irreducible by [Theorem 5.3.1](#). Over  $\mathbb{F}_2$ :  $2 = 0$ , so  $x^2 + 2 = x \cdot x$  is reducible. Over  $\mathbb{F}_3$ :  $2 = -1$ , so  $x^2 + 2 = x^2 - 1 = (x - 1)(x + 1)$  is reducible.
- (b) Over  $\mathbb{F}_2$ : the values at 0 and 1 are 1 and  $3 = 1$ , so there is no root, and the cubic is irreducible by [Theorem 5.3.1](#). Over  $\mathbb{F}_3$ : the value at 1 is  $3 = 0$ , so it is reducible; indeed  $x^3 + x + 1 = (x - 1)(x^2 + x + 2)$  in  $\mathbb{F}_3[x]$ , as expanding shows:  $x^3 + x^2 + 2x - x^2 - x - 2 = x^3 + x - 2 = x^3 + x + 1$ .
- (c) By [Theorem 5.3.6](#), a rational root  $u/v$  in lowest terms has  $u \mid -1$  and  $v \mid 1$ , so it is 1 or  $-1$ . The values are  $1 - 3 - 1 = -3$  and  $-1 + 3 - 1 = 1$ . No rational root, so the cubic is irreducible over  $\mathbb{Q}$  by [Theorem 5.3.1](#).
- (d) Reducible, although it has no rational root (its values are at least 4): expanding,  $(x^2 + 2x + 2)(x^2 - 2x + 2) = x^4 - 4x^2 + 4 + 4x^2 = x^4 + 4$ . This is the warning after [Theorem 5.3.1](#) in action.
- (e) Irreducible by [Theorem 5.3.7](#) with  $p = 3$ :  $3 \nmid 1$ ; 3 divides the coefficients 0, 0, 0,  $-6$ , 3 of  $x^4, x^3, x^2, x, 1$ ; and  $9 \nmid 3$ .

### Exercise B2: Factoring $x^6 - 1$ Over $\mathbb{Q}$ .

Factor  $x^6 - 1$  into monic irreducible polynomials over  $\mathbb{Q}$ , and justify that each factor is irreducible. Hence find its factorization into monic irreducibles over  $\mathbb{F}_2$  as well. Is it obtained by reducing the coefficients of each rational factor modulo 2? Explain why, in general, this has to be checked rather than assumed.

*Solution.* Using  $x^6 - 1 = (x^3 - 1)(x^3 + 1)$  and Exercise (B2: A Factor of  $x^n - c^n$ ) (with  $c = 1$  and  $c = -1$ ),

$$x^6 - 1 = (x - 1)(x^2 + x + 1)(x + 1)(x^2 - x + 1),$$

which one checks by expanding:  $(x - 1)(x^2 + x + 1) = x^3 - 1$  and  $(x + 1)(x^2 - x + 1) = x^3 + 1$ . The linear factors are irreducible. For the quadratics,  $c^2 \pm c + 1 = (c \pm \frac{1}{2})^2 + \frac{3}{4} > 0$  for real  $c$ , so they have no rational root and are irreducible by [Theorem 5.3.1](#). By [Theorem 5.3.5](#) this is the factorization over  $\mathbb{Q}$ .

Over  $\mathbb{F}_2$ , reducing coefficients gives  $(x + 1)(x^2 + x + 1)(x + 1)(x^2 + x + 1)$ , since  $-1 = 1$ . Here  $x + 1$  is irreducible and  $x^2 + x + 1$  is irreducible over  $\mathbb{F}_2$  (it has no root), so the factorization over  $\mathbb{F}_2$  is  $(x + 1)^2(x^2 + x + 1)^2$ . So yes, in this case the factorization is obtained by reduction, but only because the reduced factors happen to stay irreducible, and that has to be checked: two distinct rational factors,  $x - 1$  and  $x + 1$ , became equal, and in general a factor that is irreducible over  $\mathbb{Q}$  may become reducible after reduction, as  $x^2 + 1$  does over  $\mathbb{F}_2$  in [Example \(Factoring  \$x^4 - 1\$  Over Six Fields\)](#).

### Exercise B3: Small Irreducibles Over $\mathbb{F}_2$ .

List all irreducible polynomials over  $\mathbb{F}_2$  of degree at most 3. Justify that your list is complete.

*Hint:* a polynomial over  $\mathbb{F}_2$  has the root 0 exactly when its constant term is 0, and the root 1 exactly when its number of non-zero terms is even.

*Solution.* Every non-zero polynomial over  $\mathbb{F}_2$  is monic, since its leading coefficient is 1. For  $f = \sum a_k x^k$ ,  $f(0) = a_0$  and  $f(1) = \sum a_k$ , which is 0 exactly when the number of non-zero coefficients is even.

*Degree 1:*  $x$  and  $x + 1$ , both irreducible ([Example \(First Irreducible and Reducible Polynomials\)](#) (a)).

*Degrees 2 and 3:* by [Theorem 5.3.1](#),  $f$  is irreducible exactly when  $f(0) = 1$  and  $f(1) = 1$ , that is, the constant term is 1 and the number of non-zero terms is odd. In degree 2, the polynomials with constant term 1 are  $x^2 + 1$  (two terms) and  $x^2 + x + 1$  (three terms), so only  $x^2 + x + 1$  is irreducible. In degree 3, they are  $x^3 + 1$  (two terms),  $x^3 + x + 1$  and  $x^3 + x^2 + 1$  (three terms), and  $x^3 + x^2 + x + 1$  (four terms). So the irreducible ones are  $x^3 + x + 1$  and  $x^3 + x^2 + 1$ .

Hence the complete list is  $x$ ,  $x + 1$ ,  $x^2 + x + 1$ ,  $x^3 + x + 1$ ,  $x^3 + x^2 + 1$ .

### 5.3.5.3 C. Going deeper

#### Exercise C1: Infinitely Many Irreducibles.

Let  $F$  be any field.

- (a) Prove that there are infinitely many monic irreducible polynomials over  $F$ .
- (b) Deduce that for every prime  $p$  and every  $N \in \mathbb{N}$ , there is an irreducible polynomial over  $\mathbb{F}_p$  of degree greater than  $N$ .
- (c) Explain why (a) is immediate when  $F$  is infinite, and use [Exercise \(B3: Small Irreducibles Over  \$\mathbb{F}\_2\$ \)](#) to name an irreducible polynomial of degree 3 over  $\mathbb{F}_2$ .

*Hint for (a): imitate the proof that there are infinitely many primes (Theorem 0.2.13).*

*Solution.*

- (a) Suppose, for a contradiction, that  $p_1, \dots, p_k$  are all the monic irreducible polynomials over  $F$ . The list is non-empty, since  $x$  is monic and irreducible. Let  $f = p_1 p_2 \cdots p_k + 1$ . Each  $p_i$  has degree at least 1, so  $\deg(p_1 \cdots p_k) \geq 1$  by [Theorem 0.8.3](#), and  $f$  is non-constant by [Theorem 0.8.2](#). By [Theorem 5.3.5](#) (a),  $f$  has a monic irreducible factor, which must be some  $p_j$ . Then  $p_j \mid f$  and  $p_j \mid p_1 \cdots p_k$ , so  $p_j \mid f - p_1 \cdots p_k = 1$  by [Proposition 5.1.2](#) (b). By [Proposition 5.1.2](#) (c),  $\deg p_j \leq \deg 1 = 0$ , contradicting  $\deg p_j \geq 1$ . Hence there are infinitely many monic irreducibles.
- (b) A monic polynomial of degree  $n$  over  $\mathbb{F}_p$  is determined by its  $n$  lower coefficients, so there are exactly  $p^n$  of them, and there are  $1 + p + \cdots + p^N$  monic polynomials of degree at most  $N$ , a finite number. By (a), infinitely many monic irreducibles exist, so some of them have degree greater than  $N$ .
- (c) If  $F$  is infinite, the polynomials  $x - c$  for  $c \in F$  are infinitely many distinct monic polynomials of degree 1, all irreducible by [Example \(First Irreducible and Reducible Polynomials\)](#) (a). Over a finite field there are only finitely many polynomials of degree 1, so the content of (a) is that irreducibles of higher and higher degree keep appearing. Over  $\mathbb{F}_2$ , for example,  $x^3 + x + 1$  is irreducible of degree 3 by [Exercise \(B3: Small Irreducibles Over  \$\mathbb{F}\_2\$ \)](#).

**Exercise C2: Gauss's Lemma and Eisenstein's Criterion.**

Let  $\mathbb{Z}[x]$  be as in [Theorem 5.3.6](#), and let  $p$  be a prime. For  $g = \sum b_i x^i \in \mathbb{Z}[x]$ , define its **reduction modulo  $p$**  as  $\bar{g} = \sum [b_i] x^i \in \mathbb{F}_p[x]$ , where  $[b]$  is the class of  $b$  in  $\mathbb{F}_p = \mathbb{Z}/p\mathbb{Z}$  (Example (Arithmetic on  $\mathbb{Z}/n\mathbb{Z}$  and the Field  $\mathbb{F}_2$ )).

- (a) Prove that  $\overline{g+h} = \bar{g} + \bar{h}$  and  $\overline{gh} = \bar{g}\bar{h}$  for all  $g, h \in \mathbb{Z}[x]$ , and that  $\deg \bar{g} \leq \deg g$ .
- (b) Let  $g, h \in \mathbb{Z}[x]$ , and suppose  $p$  divides every coefficient of  $gh$ . Prove that  $p$  divides every coefficient of  $g$  or every coefficient of  $h$ .
- (c) (**Gauss's lemma.**) Let  $f \in \mathbb{Z}[x]$ , and suppose  $f = gh$  with  $g, h \in \mathbb{Q}[x]$ . Prove that there is a non-zero  $\lambda \in \mathbb{Q}$  with  $\lambda g \in \mathbb{Z}[x]$  and  $\lambda^{-1}h \in \mathbb{Z}[x]$ .
- (d) Let  $F$  be a field,  $c \in F$  non-zero,  $n \in \mathbb{N}$ , and  $u, v \in F[x]$  with  $uv = cx^n$ . Prove that  $u = \beta x^s$  and  $v = \gamma x^t$  for some non-zero  $\beta, \gamma \in F$  and  $s, t \in \mathbb{N}$ .
- (e) Prove [Theorem 5.3.7](#).

*Hint for (c): among all ways of writing  $kf = (\alpha g)(\beta h)$  with  $\alpha, \beta \in \mathbb{Q}$ ,  $\alpha\beta = k$  a positive integer and  $\alpha g, \beta h \in \mathbb{Z}[x]$ , take  $k$  as small as possible. Hint for (e): suppose  $f = gh$  with  $g, h$  non-constant, use (c), and reduce modulo  $p$ .*

*Solution.*

- (a) By Example (Arithmetic on  $\mathbb{Z}/n\mathbb{Z}$  and the Field  $\mathbb{F}_2$ ),  $[a] + [b] = [a + b]$  and  $[a][b] = [ab]$ . Let  $g = \sum b_i x^i$  and  $h = \sum c_i x^i$ . The coefficient of  $x^k$  in  $g + h$  is  $b_k + c_k$ , whose class is  $[b_k] + [c_k]$ , the coefficient of  $x^k$  in  $\bar{g} + \bar{h}$ . The coefficient of  $x^k$  in  $gh$  is  $\sum_{i+j=k} b_i c_j$ , whose class is  $\sum_{i+j=k} [b_i][c_j]$ , the coefficient of  $x^k$  in  $\bar{g}\bar{h}$  (Definition 0.8.2). Finally, if  $b_i = 0$  then  $[b_i] = 0$ , so  $\bar{g}$  has no non-zero coefficient beyond position  $\deg g$ .
- (b) The hypothesis says  $\overline{gh} = 0$ . By (a),  $\bar{g}\bar{h} = 0$  in  $\mathbb{F}_p[x]$ . Since  $p$  is prime,  $\mathbb{F}_p$  is a field (Theorem 0.7.3), so Corollary 0.8.4 gives  $\bar{g} = 0$  or  $\bar{h} = 0$ . That is,  $p$  divides every coefficient of  $g$  or every coefficient of  $h$ .
- (c) Choose positive integers  $m, n$  with  $mg \in \mathbb{Z}[x]$  and  $nh \in \mathbb{Z}[x]$ , for instance the products of the denominators of the coefficients. Let  $K$  be the set of positive integers  $k$  for which there are  $\alpha, \beta \in \mathbb{Q}$  with  $\alpha\beta = k$ ,  $\alpha g \in \mathbb{Z}[x]$  and  $\beta h \in \mathbb{Z}[x]$ . Then  $mn \in K$ , so by Theorem 0.2.11  $K$  has a least element  $k$ , with corresponding  $\alpha, \beta$ . Suppose  $k > 1$ . By Proposition 0.2.12, some prime  $q$  divides  $k$ . Now  $(\alpha g)(\beta h) = \alpha\beta gh = kf$ , and every coefficient of  $kf$  is divisible by  $q$ . By (b) with the prime  $q$ ,  $q$  divides every coefficient of  $\alpha g$  or every coefficient of  $\beta h$ ; say the first (the other case is symmetric). Then  $(\alpha/q)g \in \mathbb{Z}[x]$ ,  $\beta h \in \mathbb{Z}[x]$ , and  $(\alpha/q)\beta = k/q$  is a positive integer smaller than  $k$ , which lies in  $K$ . This contradicts the minimality of  $k$ . Hence  $k = 1$ , so  $\beta = \alpha^{-1}$ , and  $\lambda = \alpha$  works.
- (d) Since  $uv \neq 0$ , both  $u, v$  are non-zero. Write  $u = \sum u_i x^i$ ,  $v = \sum v_j x^j$ , let  $i_0$  and  $j_0$  be the smallest indices with  $u_{i_0} \neq 0$  and  $v_{j_0} \neq 0$ , and let  $s = \deg u$ ,  $t = \deg v$ . The coefficient of  $x^{i_0+j_0}$  in  $uv$  is  $\sum_{i+j=i_0+j_0} u_i v_j$ . A term with  $i < i_0$  has  $u_i = 0$ , and a term with  $i > i_0$  has  $j < j_0$ , so  $v_j = 0$ . So this coefficient is  $u_{i_0} v_{j_0} \neq 0$ . The only non-zero coefficient of  $cx^n$  is at  $x^n$ , so  $i_0 + j_0 = n$ . By Theorem 0.8.3, also  $s + t = n$ . Since  $i_0 \leq s$  and  $j_0 \leq t$ , this forces  $i_0 = s$  and  $j_0 = t$ . So  $u$  has exactly one non-zero coefficient,  $u = u_s x^s$ , and likewise  $v = v_t x^t$ .
- (e) Suppose  $f$  is reducible over  $\mathbb{Q}$ :  $f = gh$  with  $g, h \in \mathbb{Q}[x]$ ,  $\deg g = s \geq 1$ ,  $\deg h = t \geq 1$ , and  $s + t = n$ . By (c), replacing  $g, h$  by  $\lambda g, \lambda^{-1}h$  (which have the same

degrees), we may assume  $g, h \in \mathbb{Z}[x]$ . Write  $g_0, h_0$  for their constant coefficients, so  $a_0 = g_0 h_0$ . Reduce modulo  $p$ . By conditions (1) and (2),  $\bar{f} = [a_n]x^n$  with  $[a_n] \neq 0$ . By (a),  $\bar{g}\bar{h} = [a_n]x^n$ , so by (d),  $\bar{g} = \beta x^{s'}$  and  $\bar{h} = \gamma x^{t'}$  with  $\beta, \gamma \neq 0$ . By Theorem 0.8.3,  $s' + t' = n = s + t$ , while  $s' \leq s$  and  $t' \leq t$  by (a). Hence  $s' = s \geq 1$  and  $t' = t \geq 1$ . So the constant coefficients of  $\bar{g}$  and  $\bar{h}$  are 0, that is,  $p \mid g_0$  and  $p \mid h_0$ . Then  $p^2 \mid g_0 h_0 = a_0$ , contradicting condition (3). Therefore  $f$  is irreducible over  $\mathbb{Q}$ ; it is non-constant since  $n \geq 1$ .